

IT-IM 05

DATA WAREHOUSING AND DATA MINING

UNIT 2 – SUPPLEMENT – NORMALIZATION REVIEW

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SUBJECT TEACHER



Normalization

- is the process of identifying and structuring the attributes in order to minimize or avoid data redundancies and anomalies.
- comes in series of forms:
 - ☐ First Normal Form (1NF)
 - ☐ Second Normal Form (2NF)
 - ☐ Third Normal Form (3NF)

Normalization



- Main objective of normalization is to avoid anomalies
 - ☐ Insertion anomaly
 - ☐ Update anomaly
 - ☐ Deletion anomaly

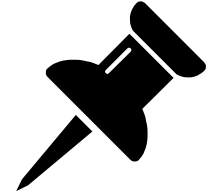
First Normal Form (1NF)



- Characteristics:

- ☐ No multi-valued attribute
- ☐ No repeating groups
- ☐ Primary key / key attributes are identified
- ☐ Others:
 - ☐ Attribute domain should not change
 - ☐ Unique names for columns and attributes

Example 1 (1NF)



Unnormalized

TeacherNo	TeacherName	Subject
1	A	CC101, CC100
2	B	CC100, ITNET01

1NF

TeacherNo	TeacherName	Subject
1	A	CC101
1	A	CC100
2	B	CC100
2	B	ITNET01

<u>TeacherNo</u>	TeacherName	Subject
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Second Normal Form (2NF)



- Characteristics:
 - ☐ Must be in 1NF
 - ☐ No partial dependencies
 - ☐ Partial dependencies exist a non key attribute is dependent only on a portion of the primary key



Example (2NF)

Suppose we have this table in 1NF. How do we convert it into 2NF?

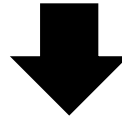
<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept	Dean	<u>SubjectCode</u>	SubjectDesc	Units	<u>SchedNo</u>	Time	Day	Type
1	A	Cab City	CICT	Dr. IM	CC101	Computer Programming Fundamentals	3	1	9:00 – 11:00	Monday	Lec
1	A	Cab City	CICT	Dr. IM	CC100	Introduction to Computing	3	2	9:00 – 11:00	Tuesday	Lec
2	B	Cab City	CICT	Dr. IM	CC100	Introduction to Computing	3	3	1:00 – 3:00	Monday	Lec
2	B	Cab City	CICT	Dr. IM	ITNET01	Introduction to Networks	3	4	8:00 – 10:00	Tuesday	Lec

Example (2NF)



1NF – Teacher Table

<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept	Dean	<u>SubjectCode</u>	SubjectDesc	Units	<u>SchedNo</u>	Time	Day	Type
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TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept	Dean
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SUBJECT

<u>SubjectCode</u>	SubjectDesc	Units
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TEACHING_SCHEDULE

<u>SchedNo</u>	Time	Day	Type
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Example (2NF)



From here, we can also identify the foreign keys that are responsible for referential integrity.

In this example, Teaching_Schedule can use TeacherNo and SubjectCode to be able to identify what subject the schedule is for and who will be teaching it

TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept	Dean
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SUBJECT

<u>SubjectCode</u>	SubjectDesc	Units
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TEACHING_SCHEDULE

<u>SchedNo</u>	Time	Day	Type	TeacherNo	SubjectCode
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Third Normal Form (3NF)



- Characteristics:
 - ☐ Must be in 2NF
 - ☐ No transitive dependencies
 - ☐ Transitive dependencies exist when a non key attribute is determining another non key attribute

Example (3NF)



TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept	Dean
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Dept is a non-key attribute that determines Dean which is another non-key attribute

SUBJECT

<u>SubjectCode</u>	SubjectDesc	Units
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TEACHING_SCHEDULE

<u>SchedNo</u>	Time	Day	Type	TeacherNo	SubjectCode
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Example (3NF)



TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	Dept
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SUBJECT

<u>SubjectCode</u>	SubjectDesc	Units
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TEACHING_SCHEDULE

<u>SchedNo</u>	Time	Day	Type	TeacherNo	SubjectCode
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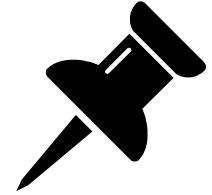
To remove the transitive dependency, we will create a new table Department.

Department

<u>Dept</u>	Dean
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Dept as foreign key in Teacher table

Example (3NF)



Scenario: What if there's a sudden change and it has been brought up that there can be a possible anomaly. Hence, your team wants to set something more unique such as DeptNo as PK.

If you do that, you need to double check if it satisfies 1NF up until your 3NF. If no conflict, update the Teacher table too and make the foreign key as DeptNo

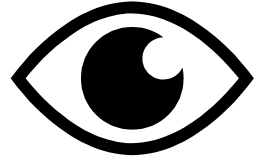
DEPARTMENT

<u>DeptNo</u>	Dept	Dean
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TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	DeptNo
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Final output



TEACHER

<u>TeacherNo</u>	TeacherName	TeacherAddress	DeptNo
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DEPARTMENT

<u>DeptNo</u>	Dept	Dean
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SUBJECT

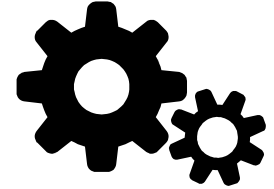
<u>SubjectCode</u>	SubjectDesc	Units
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TEACHING_SCHEDULE

<u>SchedNo</u>	Time	Day	Type	TeacherNo	SubjectCode
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From this final output, we can now create the relational schema with normalized tables.

Summary



1NF

No multi valued
attributes

2NF

No partial
dependencies

3NF

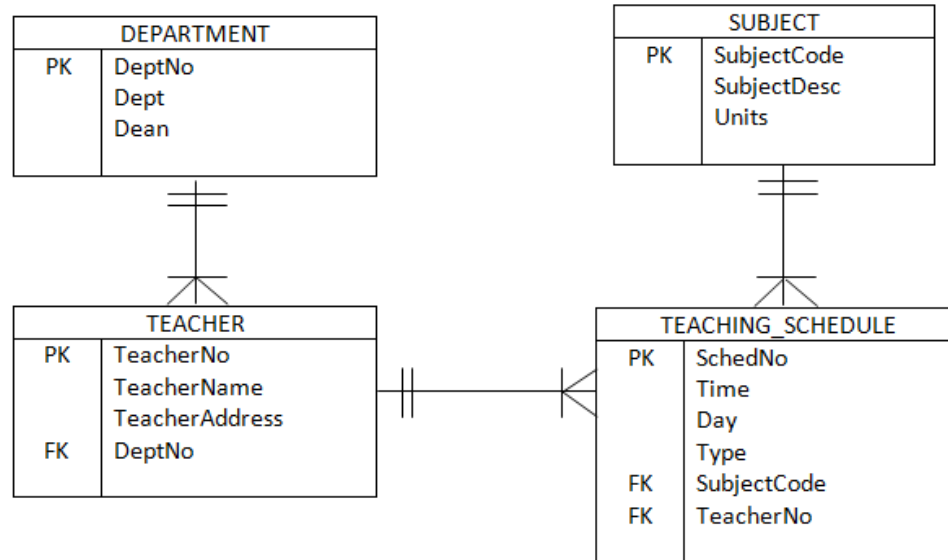
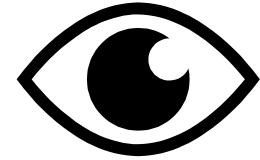
No transitive
dependencies

Relational Schema



- A **schema** is a logical grouping of database objects related to one another and becomes a guide on how the database is constructed.
- It is the blueprint of a database

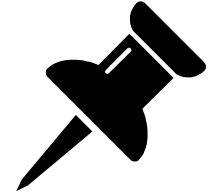
Schema



Per our schema:

- ✓ Each table has primary key which is unique across all records
- ✓ Related tables are connected through the foreign keys
- ✓ One-to-many relationships exist in tables

Schema to Table



Department

<u>DeptNo</u>	Dept	Dean
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Subject

<u>SubjectCode</u>	SubjectDesc	Units
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Assume that we did some coding to create database and tables, from the model/schema, it will become these sample table structures.

Schema to Table



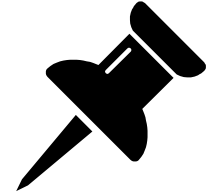
Teacher

<u>TeacherNo</u>	TeacherName	TeacherAddress	DeptNo
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Teaching Schedule

<u>SchedNo</u>	Time	Day	Type	SubjectCode	TeacherNo
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Schema to Table



Department

<u>DeptNo</u>	Dept	Dean
1	A	Dr. Dean

Subject

<u>SubjectCode</u>	SubjectDesc	Units
CC104	Information Management	3

If we are going to visualize the table with sample contents, we have to consider the columns we have defined as keys.

Schema to Table



Teacher

<u>TeacherNo</u>	TeacherName	TeacherAddress	DeptNo
123	B	Cabanatuan City	1

Teaching Schedule

<u>SchedNo</u>	Time	Day	Type	SubjectCode	TeacherNo
1	10:00 – 12:00	Monday	Lecture	CC104	123

Notice that the primary key DeptNo from Department table is a foreign key in the Teacher table.

Following the concept of foreign keys, the value of DeptNo in the Teacher table should be something that exists in the Department table.

Same goes for the TeachingSchedule and its foreign keys.

The SubjectCode and TeacherNo values should exist in their respective tables before you can input them in the TeachingSchedule table.

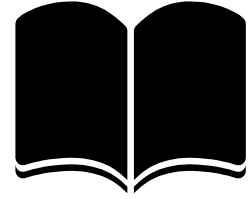
Supplemental Videos



- For more information about normalization, here are some suggested videos:

- ☐ First Normal Form: <https://www.youtube.com/watch?v=jgUeOjImOOw&app=desktop>
- ☐ Second Normal Form: <https://www.youtube.com/watch?v=9L10Q1nAfyg>
- ☐ Third Normal Form: https://www.youtube.com/watch?v=_K7fcFQowy8

References



- CC-104 Information Management Module
- Coronel, Carlos et. al. Database Principles: Fundamentals of Design, Implementation and Management.
- Bensi, Leonylyn. Database Management System 1 Worktext
- Module on IT IM-01 – Advanced Database Systems
- Hoffer, Jeffrey et al. (2002). *Modern Database Management Sixth Edition*. Prentice Hall.
- Pratt, Philip and Last, Mary. (2011). *A Complete Guide to MySQL*. Cengage Learning.
- Study Tonight. *Normalization of Database*.
<https://www.studytonight.com/dbms/database-normalization.php>

Special Credits:

SlideCarnival for the presentation design